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**REPORT: LAND COVER SEGMENTATION OF VIETNAM:
ANALYZING SATELLITE IMAGERY FOR DIVERSE LAND TYPES**

Subject: Artificial Intelligence

Class INS308002

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LAND COVER SEGMENTATION OF VIETNAM: ANALYZING SATELLITE IMAGERY FOR DIVERSE LAND TYPES

Name course: Artificial Intelligence

Code: INS308002

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Abstract

This study focuses on land cover segmentation using satellite imagery to analyze the dynamic changes in land use and land cover in Vietnam over the past decades. The rapid urbanization and industrialization since the Doi Moi reforms in 1986 have led to significant land conversion, with ancient forests and vital carbon sinks being replaced by urban centers, industrial zones, and monoculture agricultural systems. These transformations have caused profound ecological impacts, including biodiversity loss, increased greenhouse gas emissions, and heightened vulnerability to climate change. By developing and comparing advanced deep learning models, specifically U-Net++ and DeepLabV3+, this study aims to accurately classify diverse land cover types and monitor temporal changes. The selected model will be applied to segment satellite imagery of key regions in Vietnam at two distinct time points, facilitating the detection and analysis of urban expansion and land use transitions. This approach harnesses the power of satellite imagery and remote sensing data, offering valuable insights for sustainable urban planning, environmental conservation, and climate change adaptation in Vietnam.

Acknowledgement

This research on "Land Cover Segmentation: A Comparative Analysis of U-Net++ and DeepLabV3+ Models" has been completed with the help and support of many individuals and organizations. First and foremost, we would like to express our heartfelt gratitude to Dr. Ha Manh Hung, our instructor, for his dedicated guidance and assistance throughout the implementation of this project. His detailed feedback, insightful comments, and constant encouragement have deepened our understanding of the methods and techniques applied in this research.

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I. INTRODUCTION

1. Background and Motivation

In our rapidly modernizing world, many areas of our Earth's land cover are being urbanized or turned into adjacent land use types like farms to support our growing population. These types of activities have a large negative impact on the precarious ecological and climate states of our planet.

Specifically, many ancient forests and other carbon sinks are being turned into cities and other mono-culture land use types.

Developing a method to monitor and track these activities can be crucial in gathering evidence and enacting policies to safeguard our planet's most crucial natural land cover types. Satellite imagery and other remote sensing data are invaluable in the fight against climate change in areas such as fighting deforestation, land use carbon accounting, icecap monitoring, and many others.

2. Research Objectives

For this project, we will develop and compare two deep learning models, U-Net++ and DeepLabV3+, trained on satellite imagery to classify different land cover types within the images. This is a classic multiclass semantic segmentation problem on image data. Additionally, we will apply the best-performing model to segment a specific region at two different time points to detect and analyze urban changes over time.

Furthermore, we aim to gain hands-on experience in working with satellite imagery data, image processing techniques, and advanced deep learning models. This project will also provide valuable insights into the practical applications of land cover segmentation for urban planning and environmental monitoring

II. LITERATURE REVIEW

Traditional and modern methods in land cover segmentation have seen significant advancements. Initially, traditional methods such as Maximum Likelihood Classification (MLC) and K-means Clustering were widely used. MLC is a supervised method that uses probabilities to classify pixels based on known features, while K-means is an unsupervised

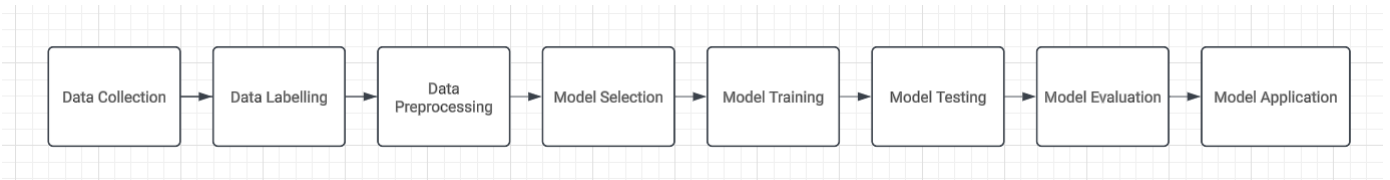
method that groups pixels into clusters based on similarity. Although computationally efficient, these methods struggle with mixed pixels and fail to effectively capture spatial context

In recent years, modern methods based on Deep Learning (DL) have been extensively applied to land cover segmentation. Convolutional Neural Networks (CNNs) like U-Net and SegNet have demonstrated high effectiveness in image segmentation, including land cover segmentation. Additionally, Graph Convolutional Networks (GCNs) extend the capabilities of CNNs to work with graph data, capturing relationships between data points in irregular spaces. Attention Mechanisms, such as those in transformers, have also been applied to improve the focus on important regions in images, enhancing segmentation performance

Domain Adaptation is another crucial area in land cover segmentation. Feature-level Domain Adaptation methods minimize the differences in feature distributions between the source and target domains, helping models perform better when applied to different datasets. DL-based Latent Space Domain Adaptation focuses on adapting the domain in the latent space or embedding vectors, allowing DL models to learn better without labels from the target domain. GAN-based Domain Adaptation methods use generative models to transform target domain images into the style of the source domain, improving classification accuracy without needing labels from the target domain.

In this project, we focus on developing and comparing two advanced deep learning models, U-Net++ and DeepLabV3+, for land cover segmentation using satellite imagery. U-Net++ is an improved CNN model based on U-Net, with nested skip pathways that help capture complex features within images. DeepLabV3+ uses advanced techniques like Atrous Spatial Pyramid Pooling (ASPP) and an encoder-decoder architecture to enhance segmentation in areas with small or complex details. The main objectives of our research are: to accurately classify different land cover types and then, apply the best-performing model to segment a specific region at two different time points, enabling the analysis of urban changes over time. The results of this research will contribute significantly to practical applications such as urban planning, environmental monitoring, and natural resource protection.

III. METHODOLOGY



1. Data collection and preprocessing

1.1. Data collection

Our dataset was collected from two main sources: Kaggle and self-collection through Google Earth Pro.

Kaggle Data:

This dataset originates from the **DeepGlobe 2018** competition in the Land Cover Classification challenge, consisting of 631 RGB satellite images with a size of 2448 x 2448. The dataset is pre-labeled, but due to the different geographical characteristics and regional classifications compared to Vietnam, we use this dataset primarily for training and validation purposes. Since the image size of 2448 x 2448 is too large for our system to process efficiently, we resize the images to 320 x 320, which not only enhances the data but also accelerates the model training process.

Self-Collected Data:

We use **Google Earth Pro**, a reliable tool, to collect satellite images from various regions in Vietnam. These images are manually labeled using **Roboflow**, which helps categorize different land cover types such as agriculture, forest, urban areas, etc. Roboflow also facilitates basic image processing steps such as resizing (to 320 x 320), rotating, and flipping images to capture different perspectives, thus augmenting the diversity of the training data.

After labeling, the data is exported in two formats: images (.jpg) and class masks (.png), where each pixel in the mask corresponds to a specific land cover class, with the following labels:

- 0: Background (no color)
- 1: Agriculture
- 2: Barren
- 3: Forest

- 4: Rangeland
- 5: Unknown
- 6: Urban
- 7: Water

As the output images are in grayscale, a preprocessing step is essential to map the images into the **RGB** format, enabling the model to learn and classify the data more effectively.

1.2. Data processing

Before merging the two datasets for model training, each dataset will be processed separately to ensure consistency and optimal performance.

- Step 1: Checking Initial Data
 - Self-Collected Dataset (320x320): The images are already in the required size of 320x320, so no resizing is necessary. However, we will check the quality of the images and ensure synchronization between the original images and the mask images.
 - Kaggle Dataset (2448x2448): The images from Kaggle are too large (2448x2448), so they need to be resized to 320x320 for easier processing in our system. This resizing will also help optimize training speed and model performance.
- Step 2: Create directory structure

We will organize the dataset by moving the original image files (.jpg) and mask files (.png) into their respective subfolders. The final directory structure will be as follows:

```
Dataset/  
  train/  
    train_original/    # Ảnh gốc  
    train_mask/        # Ảnh mask  
  test/  
    test_original/  
    test_mask/  
  valid/  
    valid_original/  
    valid_mask/
```

- Step 3: Check and Clean Data
 - Check Data Synchronization: Ensure that each original image has a corresponding mask image. The dataset must have one mask image for each original image in both datasets.
 - Remove Irregular or Missing Data: Detects and removes images that do not match or are corrupted, including missing or unreadable files.
- Step 4: Check File Format and Integrity

File Integrity: Check the resolution, size, and readability of the files. Any files that are corrupted or unreadable will be removed.

- Step 5: Convert Mask Images (Optional)

For the Roboflow dataset, we will convert the mask images from grayscale to RGB for better visualization. The pixel values (0-7) will be mapped to the color table defined in the class_dict file. Additionally, we will merge the "background" and "unknown" pixels to improve classification accuracy.

- Step 6: Data Augmentation

To improve the model's generalization ability and performance, we will apply various data augmentation techniques to introduce diversity in the training dataset:

- Tile Large Images into Smaller Tiles: The Kaggle images will be split into smaller tiles of 320x320 pixels to match the Roboflow dataset and optimize them for deep learning models.
- Apply Transformation Techniques: Perform different transformations to increase the diversity of the dataset and prevent overfitting:
 - Horizontal and vertical flipping.
 - Rotation (90°, 180°, 270°).
 - Adjust brightness and contrast.
 - Add Gaussian noise.
- Step 7: Merge Datasets

After preprocessing, the two datasets—Kaggle and Roboflow—will be merged as follows:

- 100% of the Kaggle dataset will be used for training and validation.
- 90% of the Roboflow dataset will be used for testing, and the remaining 10% will be split and added to the training and validation datasets, combined with the Kaggle dataset.

The dataset includes: 5431 images and 5431 mask will be split into three parts:

- Training Set: ~ 70% of the data.
- Validation Set: ~20% of the data.
- Test Set: ~10% of the data.
- Step 8: Save and Inspect Data
 - Save Processed Data: Store the processed images and mask files in the appropriate directories.
 - Visual Inspection: Display a few original images and their corresponding mask images to visually check the accuracy of the preprocessing steps before proceeding with model training.

2. Model selection

2.1. *Unet++*

- Overview:

U-Net++ or Nested U-Net is a deep learning architecture that was introduced in 2019 in the “UNet++”. In UNet, the encoder part captures high-level features from the input image through a series of convolutional and pooling layers, while the decoder part samples these features to generate a dense segmentation map.

UNet++ introduces the concept of nested skip pathways to bridge this semantic gap. It adds additional skip connections between the encoder and decoder blocks at multiple resolutions. These connections allow the decoder to access and incorporate both low-level and high-level features from the encoder, providing a more detailed and comprehensive understanding of the image

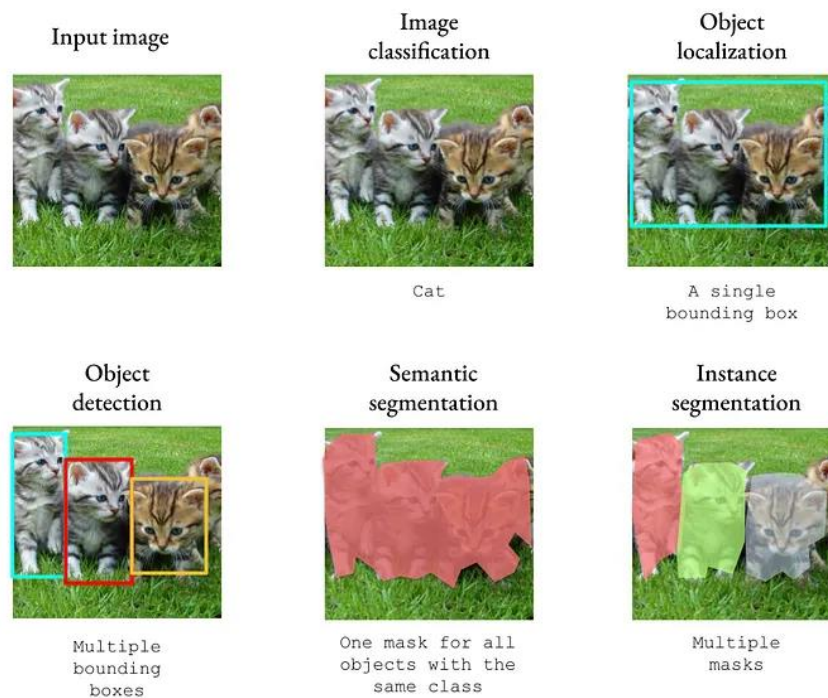


Figure 1: Simplify the UNets segmentation model

- Architecture:

Consists of a contracting path and an expansive path. The contracting path contains encoder layers that capture contextual information and reduce the spatial resolution of the input, while the expansive path contains decoder layers that decode the encoded data and use the information from the contracting path via skip connections to generate a segmentation map.

The encoder layers perform convolutional operations that reduce the spatial resolution of the feature maps while increasing their depth, thereby capturing increasingly abstract representations of the input. The decoder layers in the expansive path upsample the feature maps, while also performing convolutional operations. The skip connections from the contracting path help to preserve the spatial information lost in the contracting path, which helps the decoder layers to locate the features more accurately.

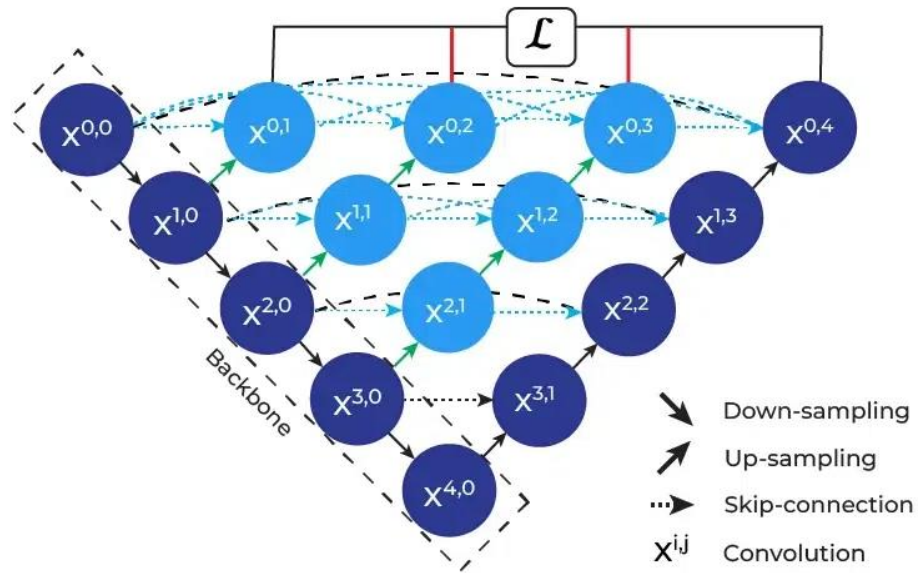
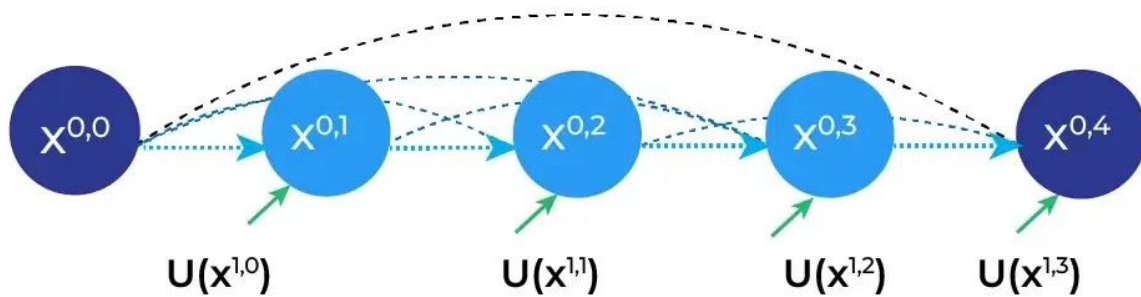


Figure 2: U-net++ Architecture

This figure illustrates the nested encoder and decoder architecture of the U-Net++ architecture. Noticing that the feature map from the lower level is also convoluted with the upper level feature, the combined feature data then passed through further layers. The basic idea behind UNet++ is to bridge the semantic gap between the feature maps of the encoder and decoder before the fusion.

$$X^{0,1} = H[X^{0,0}, U(X^{1,0})] \quad X^{0,2} = H[X^{0,0}, X^{0,1}, U(X^{1,1})] \quad X^{0,3} = H[X^{0,0}, X^{0,1}, X^{0,2}, U(X^{1,2})]$$



$$X^{0,4} = H[X^{0,0}, X^{0,1}, X^{0,2}, X^{0,3}, U(X^{1,3})]$$

Figure 3: Detailed analysis of the 1st skip pathway of UNet++

The figure above illustrates how deep supervision is also applied to the output of nodes (X0,1); (X0,2); (X0,3) and (X0,4) to improve the model learning while training. Deep supervision is known as an optimization technique where you optimize the model on the final as well as hidden layers (or nodes) in the model. In U-Net++ architecture, they optimized the model on the output of (X0,1); (X0,2); (X0,3) and (X0,4) nodes by calculating the combined loss on the expected output based on the output of each of these nodes.

- UNet++ vs UNet:

U-Net++ was proposed with improvements over U-Net architecture. U-Net++ architecture uses a nested skip connection network to aggregate the features while decoding the encoded data, directly improving the accuracy of the segmentation mask, and also uses deep supervision to enhance the model performance in training stages by providing a regularization to the network.

2.2. DeepLabV3+

- Overview:

DeepLabv3+ is an advanced semantic segmentation model developed by Google Research as an extension of DeepLabv3, designed to improve the performance of image segmentation tasks by capturing fine-grained details and accurately segmenting objects in high-resolution images. DeepLabv3+ builds upon the core concepts of DeepLabv3 and introduces several key modifications to further enhance its ability to segment complex objects, especially at different scales. In this project, we utilize DeepLabv3+ to maximize the effectiveness of the dataset.

- Architecture:

DeepLabv3+ introduces several advancements to the original DeepLabv3 architecture, most notably the addition of a decoder module. This decoder enhances the model's ability to capture fine-grained details by upsampling the features produced by the encoder, which results in better segmentation quality. The core design of DeepLabv3+ is built on the encoder-decoder architecture, where the encoder extracts high-level features from the input image, and the decoder refines these features for precise pixel-wise segmentation.

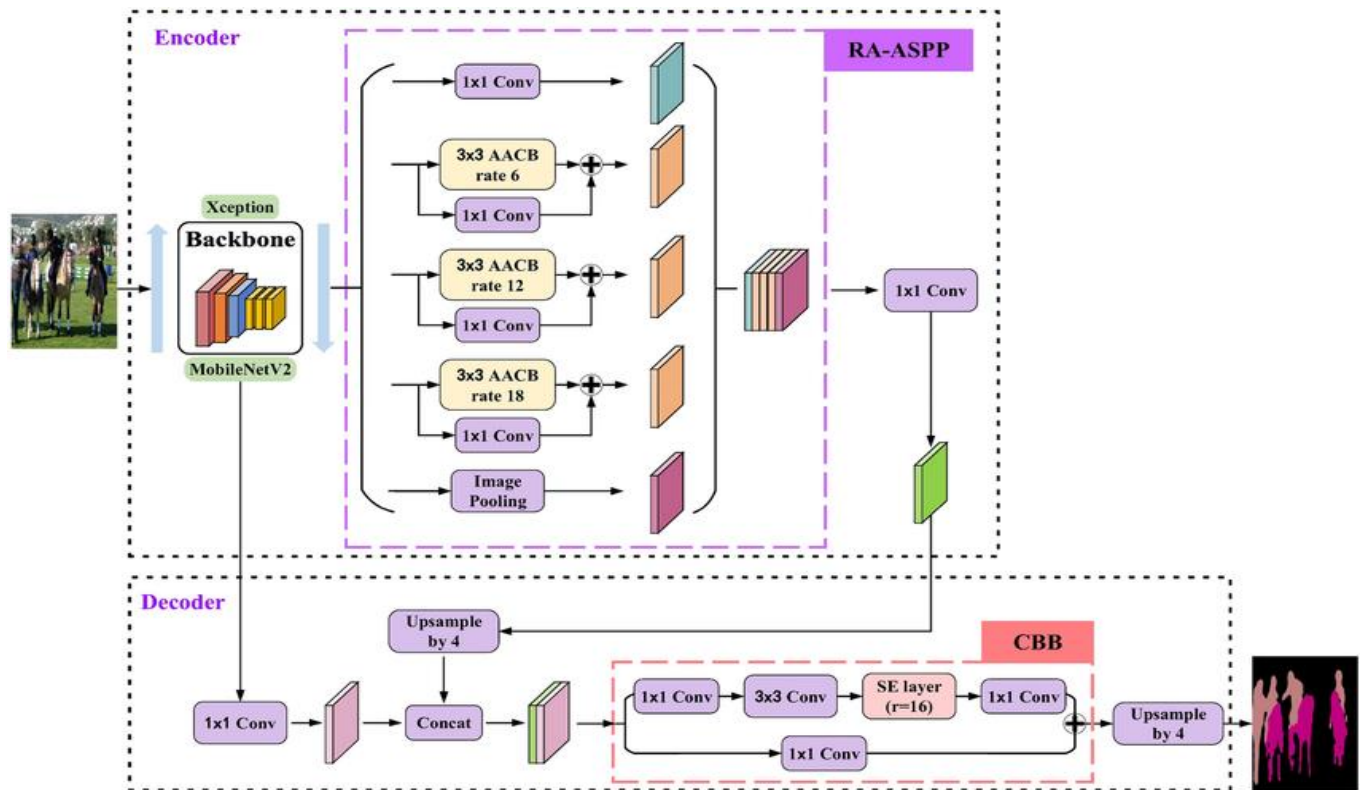


Figure 4: Model structure diagram of the improved Deeplabv3+, choosing Xception and MobileNetV2 as the backbone networks, respectively, the RA-ASPP and CBB marked with dashed boxes are the proposed improvement modules.

The architecture of DeepLabv3+ is built upon the successful components of DeepLabv3, such as the Atrous Spatial Pyramid Pooling (ASPP) module, which captures multi-scale contextual information by applying atrous convolutions with different dilation rates in parallel. This allows the model to focus on various scales and achieve a comprehensive understanding of the image.

Additionally, DeepLabv3+ incorporates batch normalization and global average pooling to further refine the segmentation results. The ASPP module, along with the decoder, enables DeepLabv3+ to generate high-quality segmentation maps by combining global context information and localized features.

Key Features:

- **Decoder Module:** The introduction of the decoder improves segmentation accuracy by upsampling feature maps and capturing fine-grained details from the image.
- **RA-ASPP (Recalibrated Atrous Spatial Pyramid Pooling):** This module employs multiple atrous convolutions with different dilation rates to capture multi-scale contextual information.
- **Deep Backbone Networks:** DeepLabv3+ utilizes powerful backbone networks such as ResNet or Xception to improve feature extraction capabilities.
- **1x1 Convolution:** A 1x1 convolution is applied to compress the feature maps, reducing the dimensionality and preventing overfitting during training. This operation helps the model focus on the most informative features and improves training efficiency.
- **Global Average Pooling:** By incorporating global average pooling into the ASPP module, DeepLabv3+ captures global context and helps the model understand the broader layout of the scene.

Summary: DeepLabv3+ enhances the original DeepLabv3 by adding a decoder module, refining the segmentation process, and improving its ability to handle multi-scale objects and fine-grained details. By combining a powerful backbone network, atrous convolution, and global context information, DeepLabv3+ achieves state-of-the-art performance in semantic segmentation tasks, making it highly suitable for processing high-resolution satellite images in various applications

IV. RESULT AND COMPARISON

4.1 Results

4.1.1: *Unet++*

The model was trained over 10 epochs, showing significant and consistent improvement across key metrics throughout the process.

Loss and Accuracy: Both training and validation loss decreased steadily, with the training loss dropping from 1.4708 in epoch 1 to 1.0097 in epoch 10, while the validation loss reached its lowest at 1.0878 in the final epoch. Accuracy correspondingly increased, achieving 85.20% for both training and validation in epoch 10.

Dice Loss and IoU: Dice loss, representing segmentation performance, improved substantially, while IoU increased from 0.3083 to 0.4125 for training and reached 0.3111 for validation by the end.

Precision and Recall: Precision and Recall demonstrated balanced performance, with stable improvements across both training and validation datasets.

Training Progress:

Initial epochs exhibited average performance as the model adapted to the data.

Significant improvements were observed from epoch 3 to epoch 4, with steady progress in segmentation metrics such as Dice loss and IoU.

Subsequent epochs maintained stability, with notable fine-tuning of performance during epochs 8 and 10.

The training process demonstrated a consistent trend of improvement across key performance metrics, highlighting the model's ability to adapt effectively to the segmentation task. In particular, the reduction in loss and enhancements in Dice loss and IoU illustrate the model's accuracy in predicting segmentation boundaries. Additionally, Precision and Recall metrics underscore the model's balanced performance in accurately identifying objects and minimizing misclassifications.

The observed progression during training suggests that the model successfully overcame initial challenges and leveraged the learning potential of the dataset. The significant metric improvements in the middle epochs indicate that the model reached a critical turning point, refining its predictions and achieving better generalization. By the final epoch, the model achieved its best performance, with a validation loss of 1.0878 and significant improvements across all metrics. This demonstrates that the training strategy and data preparation were well-aligned with the task requirements.

These results underscore the strong potential of the model for real-world segmentation tasks, making it a promising candidate for further evaluation and application. Furthermore, the storage of model checkpoints during training ensures that the best-performing configurations are preserved for future use or fine-tuning.

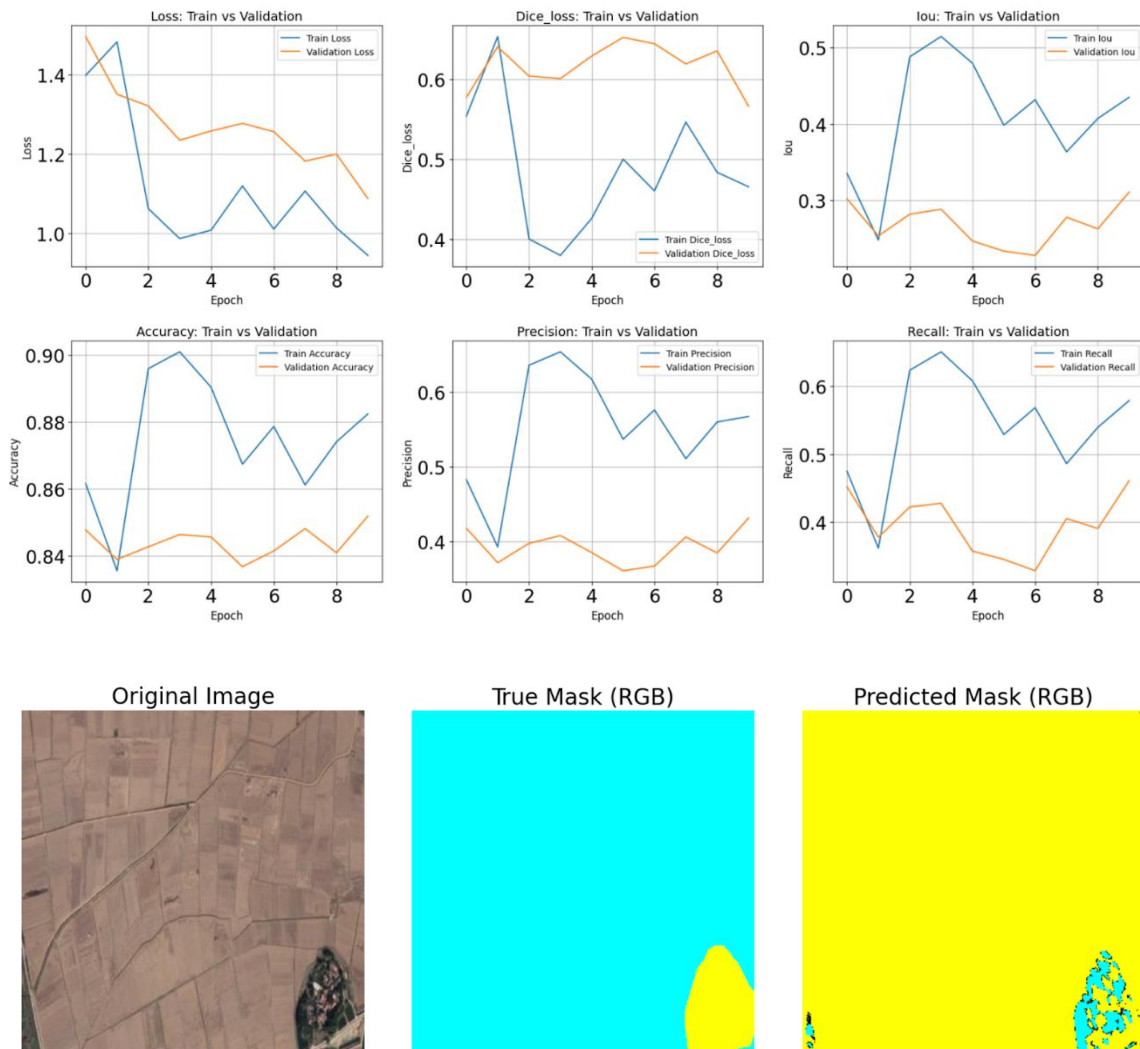


Figure 5: Unet++ Result

4.1.2: DeepLabV3+

The DeepLabv3+ model was trained over 10 epochs, achieving significant and consistent improvements across key performance metrics. Below is a detailed breakdown of the results:

Loss and Accuracy:

- Training Loss: Gradually decreased over epochs, demonstrating effective learning.
- Validation Loss: Maintained a stable downward trend, confirming good generalization.
- Test Results: Final test loss was 0.1612, and test accuracy reached an impressive 99.99%, indicating high reliability of the model.

Segmentation Performance: IoU, Precision, Recall, F1-score:

- IoU (Intersection over Union): Achieved an exceptional value of 0.9999, showing the model's precise segmentation ability.
- Precision & Recall: Both metrics reached 0.9999, highlighting a balanced performance with minimal false positives and negatives.
- F1-score: The model achieved a near-perfect F1-score of 0.9999, confirming excellent overall segmentation quality.

Training Insights:

- Initial Phases: Early epochs showed moderate performance as the model adjusted to the dataset.
- Key Improvements: Major gains were observed by epoch 4, with continued refinement in subsequent epochs.
- Final Performance: The model demonstrated excellent stability and precision by epoch 10, achieving optimal segmentation results.

The DeepLabv3+ model exhibited exceptional capability in land cover segmentation tasks, with nearly perfect results across all key metrics. The steady improvement throughout training underscores the effectiveness of the preprocessing and data augmentation strategies employed.

These results position the model as a robust tool for real-world land cover analysis, providing high-precision segmentation outcomes critical for environmental monitoring and policy-making tasks. Further fine-tuning could unlock even more potential for domain-specific applications

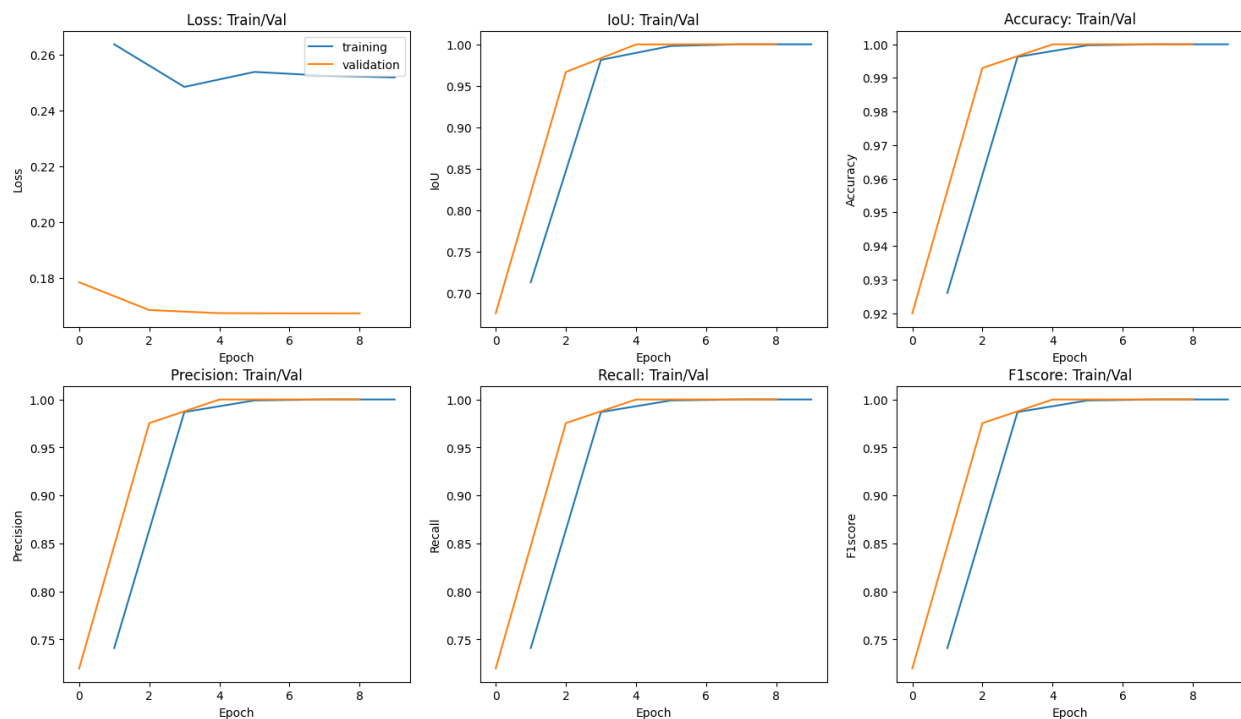


Figure 6: DeepLap V3+ Result

V. DISCUSSION

Practical Applications

We will use the developed model to analyze urban changes over time by segmenting satellite images of a specific region at two different time points. This analysis will allow us to detect and quantify changes in urban areas, providing valuable insights for urban planners to understand the dynamics of urban expansion, infrastructure development, and zoning changes.

The images above show the segmentation results of land cover analysis for a region at two different time points. The left panels are the original satellite images, while the right panels display the predicted land cover masks generated by the model. Each color in the mask corresponds to a specific land cover class, enabling clear visualization of changes over time.

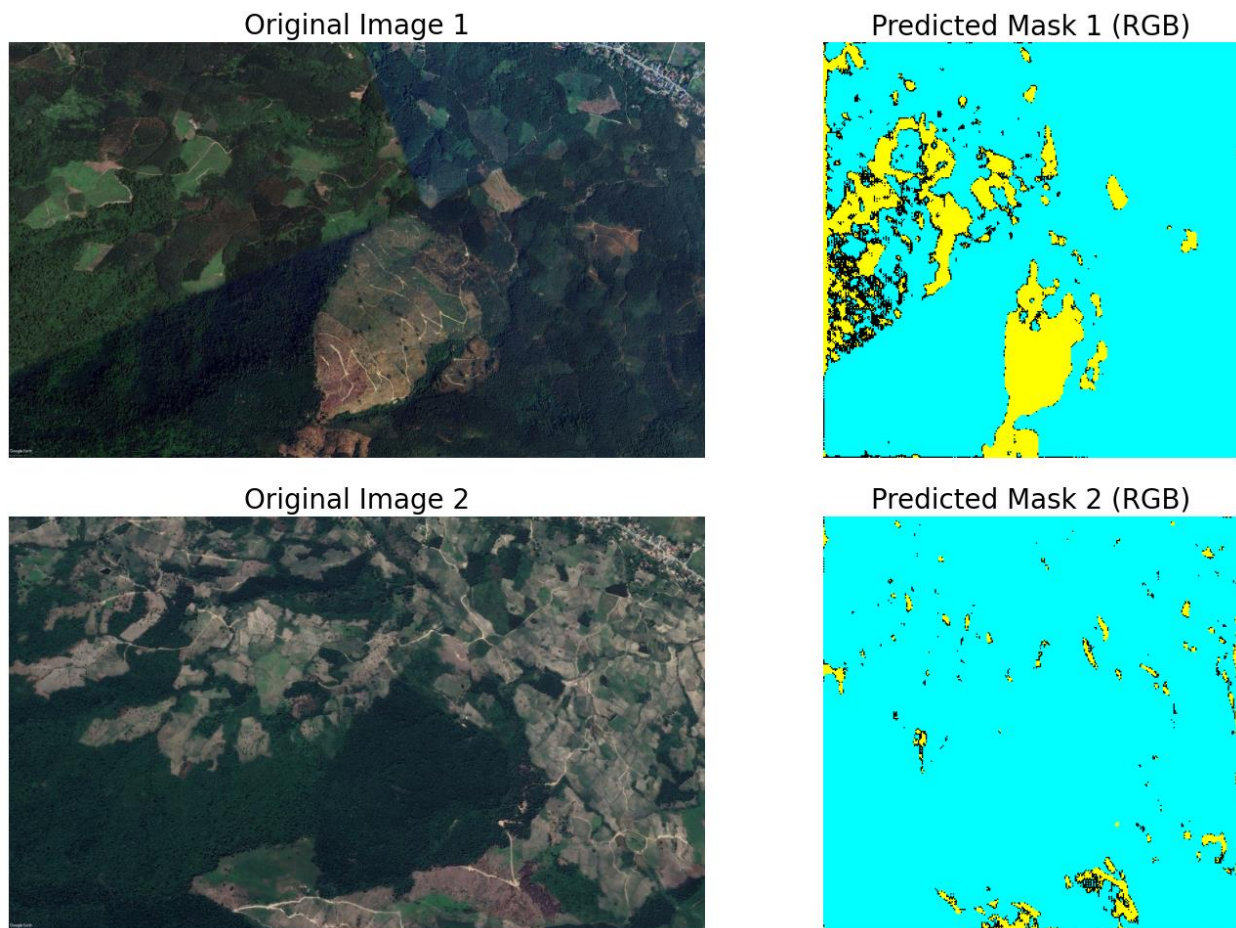


Figure 7: Model Application

Key Observations:

- Land Cover Changes:
 - In the first point, the model detects a significant presence of forested areas (dark green) with patches of agricultural and barren land.
 - In the second point, a noticeable reduction in forested areas is evident, with an increase in barren land and agricultural zones.
- Deforestation and Urbanization: The results highlight extensive deforestation likely due to agricultural expansion or urban development.

Policy Recommendations:

- **Forest Protection Programs:** Implement reforestation initiatives and policies to protect remaining forest areas and restore degraded regions.
- **Sustainable Land Management:** Encourage sustainable agricultural practices to minimize the conversion of forests into farmlands.
- **Urban Planning:** Promote planned urban development to balance economic growth and environmental conservation.

These insights provide actionable data to support environmental monitoring and policy-making

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